Nirakar Nepal

Lalitpur, Kathmandu, Nepal | +977-9866695516 | nirakarnepal71@gmail.com
linkedin.com/in/nirakar-nepal | github.com/Nirakar14

RESEARCH INTERESTS

Power system resilience and optimization under high penetration of variable renewable energy and inverter based resources(IBR) in distribution grids; nonlinear multi-objective optimization and large-scale time-series simulation for distribution network planning; condition monitoring and fault diagnosis, load flexibility and power quality in grid-interactive distribution systems; machine learning approaches for power system modeling and operational optimization and data-driven approaches for grid monitoring and resilience assessment.

EDUCATION

Bachelor of Engineering in Electrical Engineering
Institute of Engineering (IOE), Pulchowk Campus, Tribhuvan University – Nepal Apr. 2022 – 2026

- Full IOE Merit Scholarship recipient for outstanding performance in the national entrance examination

PROFESSIONAL EXPERIENCE

Electrical Engineering Intern
K&A Engineering Consulting – US-Based Firm 2026 – Present

- Conducted distribution feeder studies in CYME and PSCAD to evaluate the impact of distributed energy resource (DER) integration on grid performance, including voltage profiles, protection coordination, and thermal loading
- Contributed to power system planning and interconnection studies in the DER impact assessment department, supporting grid modernization workflows aligned with utility interconnection standards

PUBLICATIONS

1. N. Nepal, R. Upadhyay, and A. K. Panjiyar, "Physics-Based Screening and Robust Multi-Objective Optimization of Distributed Energy Resource Architectures for LV Microgrids," *Energy Conversion and Management: X* (Elsevier), vol. 31, p. 101933, 2026. doi:10.1016/j.ecmx.2026.101933
   [First Author | Published]
   Proposes a two-stage framework combining physics-based backward–forward sweep (BFS) time-series power-flow simulation, Grey Relational Analysis (GRA) screening, and robust NSGA-II multi-objective optimization to identify deployment-ready DER architectures for LV microgrids under worst-case voltage and operational constraints across 50 uncertainty scenarios.

2. N. Nepal, R. Upadhyay, S. Neupane, and U. Bhandari, "Co-Optimization of Battery Energy Storage Systems for Voltage Regulation and Reverse Power Flow Control in PV-Enriched Low Voltage Distribution Networks," *IET Smart Grid*, 2025.
   [First Author | Under Review]
   Develops a hybrid GA–PSO planning framework for joint BESS placement and sizing in high-PV LV feeders, coupled with a Dynamic Consensus Control (DCC) strategy for real-time coordinated BESS dispatch; achieves full voltage compliance (0.95–1.05 pu) and a 65% reduction in midday feeder losses on the CIGRE LV benchmark.

3. N. Nepal, R. Upadhyay, S. Neupane, U. Bhandari, and B. Silwal, "An Artificial Neural Network–Based Approach for Harmonic Component Prediction in Grid-Connected PV Systems," *Journal of Science and Engineering (JScE)*, vol. 12, no. 2, pp. 133–140, 2025. doi:10.3126/jsce.v12i2.91444

[First Author | Published | Outstanding Paper Award, IEEE RESSD 2025]
Trains a feedforward ANN on real power-quality measurements from a 6.4 kW campus PV installation to predict THD and harmonic components using solar irradiance and ambient temperature as inputs (83% accuracy), demonstrating environment-driven harmonic estimation for proactive inverter-side mitigation.

TECHNICAL SKILLS

Power Systems & Simulation	Python (power-flow, multi-period simulation), MATLAB/Simulink, CYME, PSCAD, COMSOL Multiphysics
Optimization & Control	NSGA-II, Genetic Algorithm (GA), Particle Swarm Optimization (PSO), CVXPY, Gurobi Optimizer
Programming Languages	Python, C/C++, JavaScript; currently learning Julia
ML/DL Frameworks	TensorFlow, Keras, scikit-learn, NumPy, Pandas, SciPy, Matplotlib
Research & Dev Tools	LaTeX, Git/GitHub, Jupyter Notebook

RESEARCH EXPERIENCE

Physics-Based DER Architecture Screening and Robust Multi-Objective Optimization *2024 – 2026*
IOE Pulchowk Campus, Department of Electrical Engineering

- Implemented a backward–forward sweep (BFS) power-flow solver in Python for large-scale time-series simulation of the 18-bus CIGRE LV radial benchmark feeder; built mathematical models of PV generation (NOCT temperature derating), wind turbines (Weibull-based synthetic profiles), BESS SoC dynamics, microturbine, and fuel cell units
- Designed a scenario-based evaluation framework generating 50 uncertainty scenarios via multiplicative load/PV/wind perturbation factors to stress-test 15 DER architecture candidates; aggregated multi-metric performance data into a GRA decision matrix with ζ-sensitivity and weight-sensitivity analysis to verify ranking stability
- Formulated and solved a robust NSGA-II multi-objective optimizer (population 60, 60 generations, 10 optimization + 10 independent validation scenarios) minimizing mean grid-import energy and annualized total cost while enforcing worst-case voltage, export, and SoC constraints; only the high-DER architecture (A14) passed independent robust validation
- Published in *Energy Conversion and Management: X* (Elsevier, 2026)

Co-Optimization of BESS for Voltage Regulation and Grid Resilience in LV Networks *2025 – Present*
Final Year Project, IOE Pulchowk Campus, Department of Electrical Engineering

- Formulated a hybrid GA–PSO optimization framework for coupled BESS placement (discrete bus selection, $174 = 2380$ search space) and continuous sizing (power/energy ratings), evaluated on a modified CIGRE LV feeder at 30–50% PV penetration; embedded planning-time surrogate BESS action within the optimizer for voltage feasibility screening
- Designed and numerically validated a Dynamic Consensus Control (DCC) algorithm for real-time multi-unit BESS coordination; formulated discrete-time SoC dynamics, ramp-rate and power saturation constraints, and consensus gain tuning ($\tau = 2$ h tracking time constant); verified 24-hour SoC convergence and proportionate fleet energy utilization in Python
- Optimized four-unit deployment (buses 13, 14, 15, 17; total $\sim$373 kW/$\sim$382 kWh) maintains all node voltages within 0.95–1.05 pu, reduces midday feeder losses by 65%, and cuts peak upstream export from $\sim$600 kW to $\sim$250 kW — directly relevant to distribution-level grid resilience under high renewable penetration
- Manuscript submitted to *IET Smart Grid* (under review)

ANN-Based Harmonic Prediction for Grid-Connected PV Systems *2024 – 2025*
IOE Pulchowk Campus, Department of Electrical Engineering

- Collected real power-quality measurements (THD and harmonics up to the 50th order) at the PCC of a 6.4 kW rooftop PV installation using an Elcontrol NanoVIP Cube power analyzer; aligned with 10-minute irradiance/temperature records from Nepal's Department of Hydrology and Meteorology (DHM)
- Designed, trained, and evaluated a fully connected feedforward ANN (4 hidden layers; 128–96–256–160 neurons; ReLU with dropout regularization) in TensorFlow/Keras; validated irradiance and temperature as sufficient inputs via Random Forest feature importance (58%/42%), achieving THD prediction RMSE 0.039 and MAE 0.029
- Paper received **Outstanding Paper Award** at IEEE RESSD 2025; project awarded **Best Solution** at Energy Hackathon 8.0 (2025)

CERTIFICATIONS & COURSES

Hardware in Loop Fundamentals – Hil Academy *2025*
Introduction to Typhoon HIL toolchain, basic workflow troubleshooting real-time models
View Certificate

Siemens PLC Mastery (Begineer to Advance) – Udemy *2025*
Covered begineer to advanced technical topis realted to PLC

View Certificate

HONORS & AWARDS

- **Outstanding Paper Award** — IEEE International Conference on Renewable Energy and Sustainable System Development (RESSD 2025), Kathmandu, Nepal
- **Winner** — Energy Hackathon 8.0 (2025), best solution in the power systems and renewable energy track
- **IOE Full Merit Scholarship** — awarded to top-ranking students in the national IOE entrance examination, Institute of Engineering, Tribhuvan University, Nepal

LEADERSHIP & SERVICE

President, Electrical Club — IOE Pulchowk Campus *May 2024 – May 2025*

Directed club operations for 200+ members; organized technical workshops and seminars on power systems and renewable energy; coordinated inter-departmental activities and industry speaker events.